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Printed Pages : 3

337846(37)

BE (8th Semester)

Examination, April - May, 2017

[New Scheme]

Environment Pollution and Control

Time Allowed : 3 hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (*i*) **All** questions are compulsory. Part (*a*) of each question is compulsory. Attempt any **two** parts from (*b*), (*c*) and (*d*) of each question.

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) Differentiate between primary pollutants and secondary pollutants. [2]

(*b*) What are the economic effects of air pollutants? Explain with specific example of materials damage. [7]

(*c*) Define smoke, mist and fog. What are the harmful effects of polluted air on human beings? [7]

(*d*) Write note on Aerosols. [7]

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2. (*a*) Write shortly on Exhaust Emission, Crank Case Emission. [2]

(*b*) Explain the photochemical smog formation from automobile exhaust. [7]

(*c*) Explain the sources of air pollution and method of abatement in thermal power plants. [7]

(*d*) Draw the line diagram of ferrous metallurgical operation and explain different types of pollutions emitted from ferrous metallurgical operation. [7]

3. (*a*) Define mass and volume concentrations. [2]

(*b*) Explain the principle, construction and use of the Ringelmann chart for smoke measurement. [7]

(*c*) How gaseous contaminants of combustion are controlled? [7]

(*d*) What do you mean by smoke? List the methods to control smoke with the help of diagram. Explain the working of Hartridge smoke meter. [7]

4. (*a*) What are the various means available for the control of effluent discharge into the atmosphere? [2]

(*b*) Explain the principle of electrostatic precipitator. Discuss merits, demerits and industrial application of electrostatic precipitator. [7]

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(*c*) With neat sketch, write the principle, construction and working of cyclone separator. [7]

(*d*) What are the objectives of using control equipment? Discuss the factors to be considered while selecting suitable control equipment for particulate removal. [7]

5. (*a*) Define Air Pollution Index. [2]

(*b*) What do you mean by constitution of Board? Explain the function of the Central Pollution Control and State Pollution Control Board. [7]

(*c*) Describe the phenomenon of greenhouse effect due to carbon dioxide. What are the threats of greenhouse and how it can be prevented? [7]

(*d*) Discuss the important features of Air Act 1981 and 1986. [7]

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